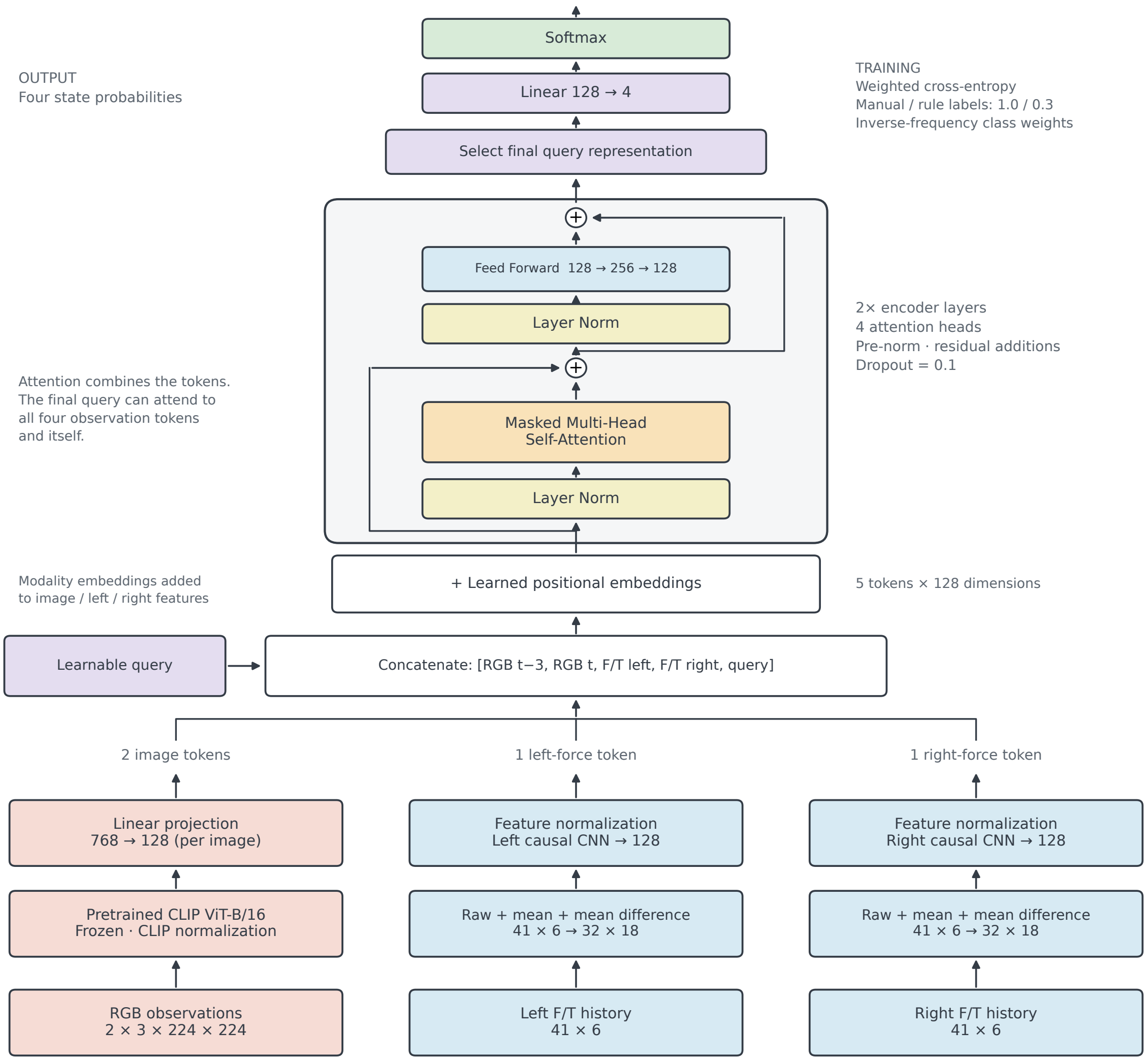
RGB–Force Context Classifier

Frozen visual backbone · Causal force histories · Transformer fusion

Approach / Turning / Recovery / Error



OUTPUT
Four state probabilities

TRAINING
Weighted cross-entropy
Manual / rule labels: 1.0 / 0.3
Inverse-frequency class weights

Attention combines the tokens. The final query can attend to all four observation tokens and itself.

2× encoder layers
4 attention heads
Pre-norm · residual additions
Dropout = 0.1

Modality embeddings added to image / left / right features

5 tokens × 128 dimensions

Shapes omit batch dimension. Only CLIP is frozen; projection, CNNs, embeddings, query and Transformer are trained.

F/T: force and torque. Mean window = 5; mean-difference lag = 5. Residual-path dropout omitted for clarity.